R13

Code No: 126ER

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
SOFTWARE TESTING METHODOLOGIES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	Write applications of Path Testing.	[2]
b)	Why is it impossible for a tester to find all the bugs in a system?	[3]
c)	Differentiate Transaction Flow and Data Flow.	[2]
d)	Why Transaction Flows are ill structured?	[3]
e)	What are nice domains?	[2]
f)	List the applications of domain testing.	[3]
g)	What is the need of path reduction?	[2]
h)	Write applications of Logic based Testing.	[3]
i)	What is graph matrix?	[2]
j)	Write Good Vs Bad state matrix.	[3]

PART - B

(50 Marks)

2.a)	What is Bug? Give Taxonomy of bugs? Discuss the consequences of bugs.	
b)	State and explain various dichotomies in software testing.	[5+5]
	OR	
3.a)	What are control and sequence bugs? How they can be caught?	
b)	Explain briefly heuristics procedures for sensitizing paths.	[5+5]
4.a)	Discuss in detail about Data - Flow testing strategies.	
b)	What is meant by Transaction Flow Testing? Explain with illustrations.	[5+5]
	OR	
5.a)	Explain heuristics procedures for sensitizing paths.	
b)	Explain data flow anomalies with examples.	[5+5]
6.a)	Discuss in detail about testability of Domains.	
b)	Describe lower path count arithmetic with example.	[5+5]
	OR	
7.a)	Discuss with example the equal - span range/Domain compatibility bugs.	
b)	State and explain various Domain Boundaries.	[5+5]

JNTUH Used papers 2023

8.a) What is a regular expression? Explain its role in flow anomaly detection with suitable examples.

b) Write about kv-chart. [5+5]

OR

9.a) What is equivalent state? Explain in detail.

b) What are some situations in which state testing may prove useful? Explain briefly. [5+5]

10.a) State and explain node reduction algorithm.

b) What is meant by power of a graph matrix? Discuss. [5+5]

OR

11.a) Write about equivalence relation and partial ordering relation.

b) What is the purpose of State testing? Write its limitations. [5+5]

---ooOoo---

JNTUH Used papers 2023